

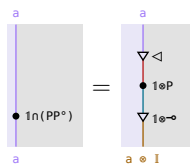
# Relation Algebra

## §5. Domain and range

(5a)

The **domain**  $\text{dom}(R) \triangleq \mathbb{1} \cap (R R^\circ)$  and the **range**  $\text{ran}(R) \triangleq \text{dom}(R^\circ)$ .

(5b)



$\triangleright$  lands the return leg back on the value  $\triangleleft$  handed out,  
so  $R^\circ \triangleright$  cuts to  $\circ$  — Frobenius

Running  $R$  and throwing the result away leaves only the fact that  $R$  could fire, and  $\text{ran}(R)$  is the same picture with the box mirrored. In  $\text{Rel}$  both steps are  $\{(a, a) : \exists b. a R b\}$ .

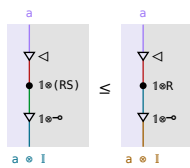
(5c)

|   |                                                                                                                                                                                                                   |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\text{dom}(R) \triangleq \mathbb{1} \cap (R R^\circ)$                                                                                                                                                            |
| 2 | $\text{Coreflexive}(X) \implies \text{dom}(R) \subseteq X \iff R \subseteq X R$                                                                                                                                   |
| 3 | $\text{dom}(R S) \subseteq \text{dom}(R)$                                                                                                                                                                         |
| 4 | $\text{dom}(R \cap S) = \mathbb{1} \cap (S R^\circ)$                                                                                                                                                              |
| 5 | $R \text{ entire} \iff \text{Entire}(R) \triangleq \text{dom}(R) = \mathbb{1}$                                                                                                                                    |
| 6 | $R \text{ simple} \iff \text{Simple}(R) \triangleq R^\circ R \subseteq \mathbb{1}$                                                                                                                                |
| 7 | $R \text{ a map} \iff \text{Map}(R) \triangleq \text{Entire}(R) \wedge \text{Simple}(R)$                                                                                                                          |
| 8 | $\text{Entire}(R) \wedge \text{Entire}(S) \implies \text{Entire}(R S)$<br>$\text{Simple}(f) \wedge \text{Simple}(g) \implies \text{Simple}(f g)$<br>$\text{Map}(f) \wedge \text{Map}(g) \implies \text{Map}(f g)$ |
| 9 | $\text{Entire}(R S) \implies \text{Entire}(R)$                                                                                                                                                                    |

### §5.1. Sliding the discard

$\text{dom}(RS) \subseteq \text{dom}(R)$ , and a single glyph for **dom** would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

(5.1a)



$S \circ \circ \sqsubseteq \circ$ , the lax axiom for  $\circ$  in the first section  
— the discard slides back past  $S$

Equality is  $S$  entire, which is the same picture read as  $\text{dom}(R) = \mathbb{1} \iff R \text{ entire}$ .